

CONSTRA

Geoscience software for engineers

Vision and Initial Scope

Draft v0.3 · 2026-04-26

*Working name: **Constra** (subject to change)*

1. Executive Summary

Constra is desktop software for engineering geologists and engineering geophysicists. It pulls together the data an engineering project actually runs on (boreholes, in-situ tests, surface geophysics, topography, CAD plans) into one project file, in one coordinate system. Map, section, and 3D views all read from the same data, so the well or the section the user picks in one view is the same well or section in the other two.

Oil and gas has had this kind of software for decades. Petrel, by Schlumberger, is the obvious example, and it does its job well. Engineering has nothing comparable. Engineering geologists draw cross-sections in CAD, plot logs in Surfer or Excel, manage spatial layers in a separate GIS package, and keep tabular records in spreadsheets. Pieces drift out of sync. Figures stop matching the data they came from. Constra is built to replace that fragmented stack with one project.

Petrel and Surfer appear in this document as reference points, not as rivals. Petrel shows what an integrated workspace can do for the geologist who uses it every day. Surfer shows where the ceiling sits for tools that only plot. Constra is for a different industry, one that today has neither.

2. Reference Point: Petrel

Petrel is Schlumberger's interpretation platform for the oil and gas industry. Inside that industry it is the standard tool, and it works well. It is reproduced here for one reason: it shows what an integrated workspace can do for the geologist who lives inside it. Engineering needs something with the same shape, just built around engineering data.

Petrel handles seismic data, well logs, geological surfaces, and reservoir models in one project. The user works in 2D maps, vertical sections, and 3D scenes, and everything stays in sync. Click a well in the map and the same well lights up in the section and the 3D scene.

2.1 Map View

The Map View shows everything in its real-world location. Satellite imagery, seismic line tracks, well heads, and cultural features (roads, buildings) layer up in the same window.

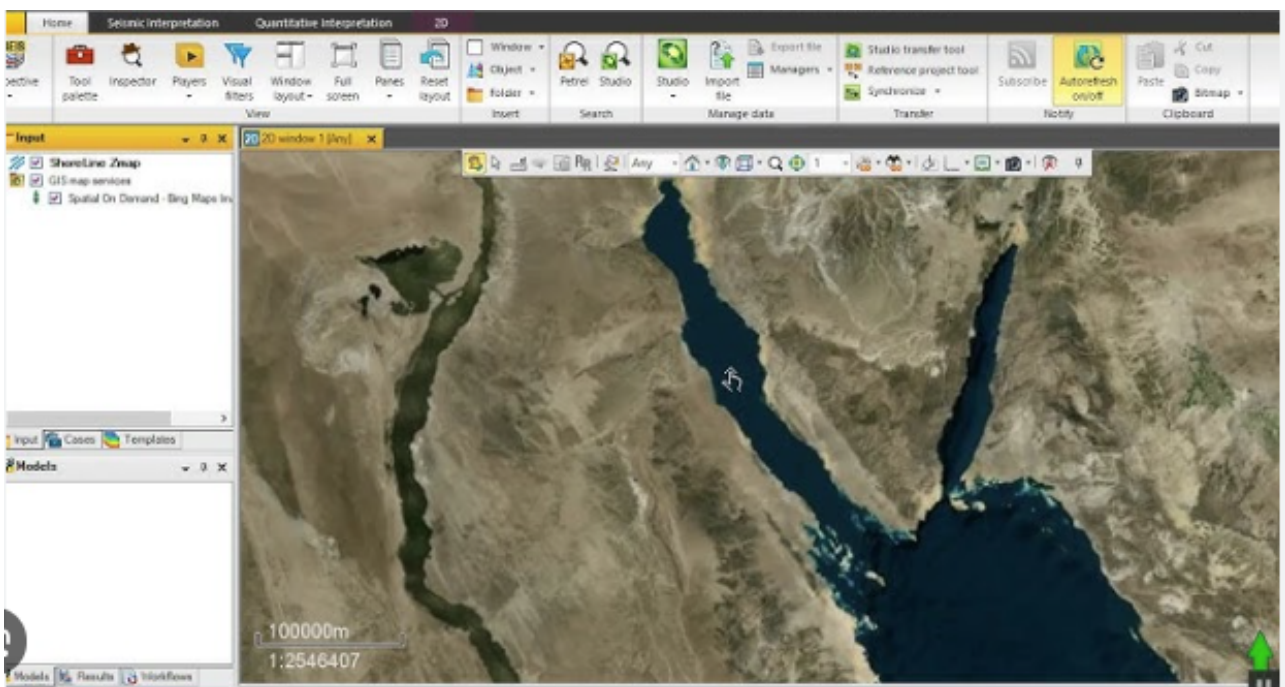


Figure 1. Satellite basemap in Petrel's Map View. The image of the project area sits at the bottom of the layer stack and gives every other layer its geographic context.

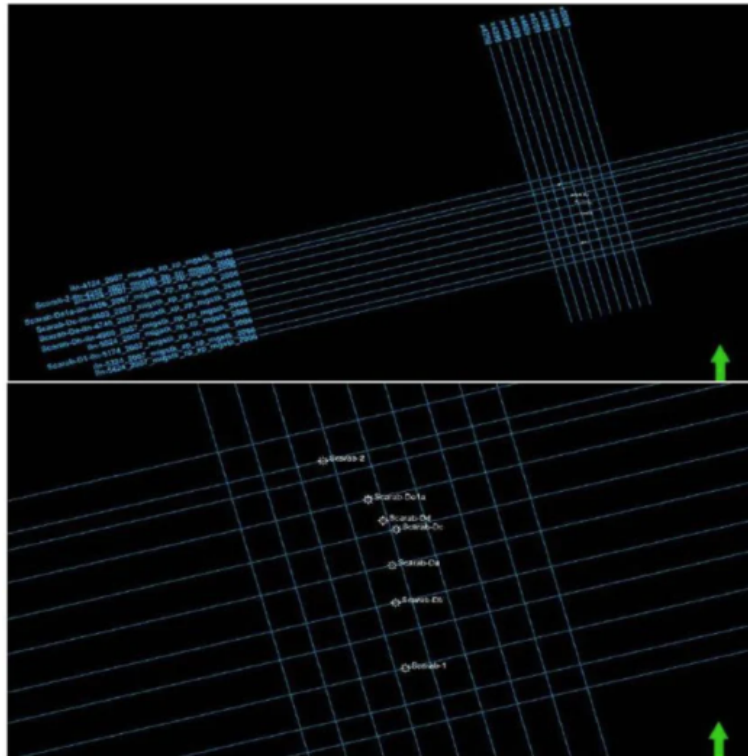


Figure 2. Seismic lines and well locations on the same map. The interpreter can see which line passes nearest to which well and open that line directly from the map.

2.2 Section View

The Section View is for data that lives on a vertical slice through the ground: 2D seismic lines, well logs, horizons, and faults. Click a line on the map and it opens here as a vertical section, ready to interpret.

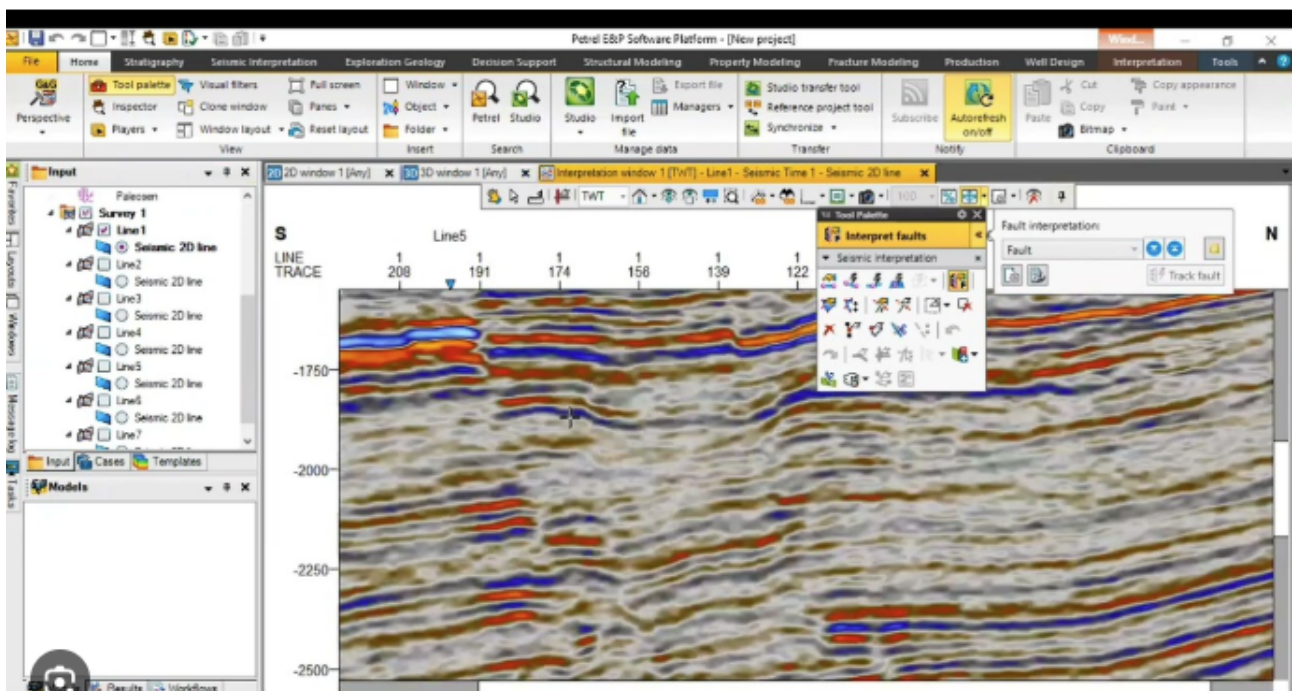


Figure 3. A 2D seismic line displayed as a vertical section in Petrel.

Map View and Section View are linked. Pick a horizon in the section and the same pick shows up on the map. Move a crosshair on the map and the section follows. The same is true for every other data type in the project.

2.3 3D View

The 3D View shows the same data in three dimensions. It is linked to the Map and Section views, so a selection made anywhere appears in all three.

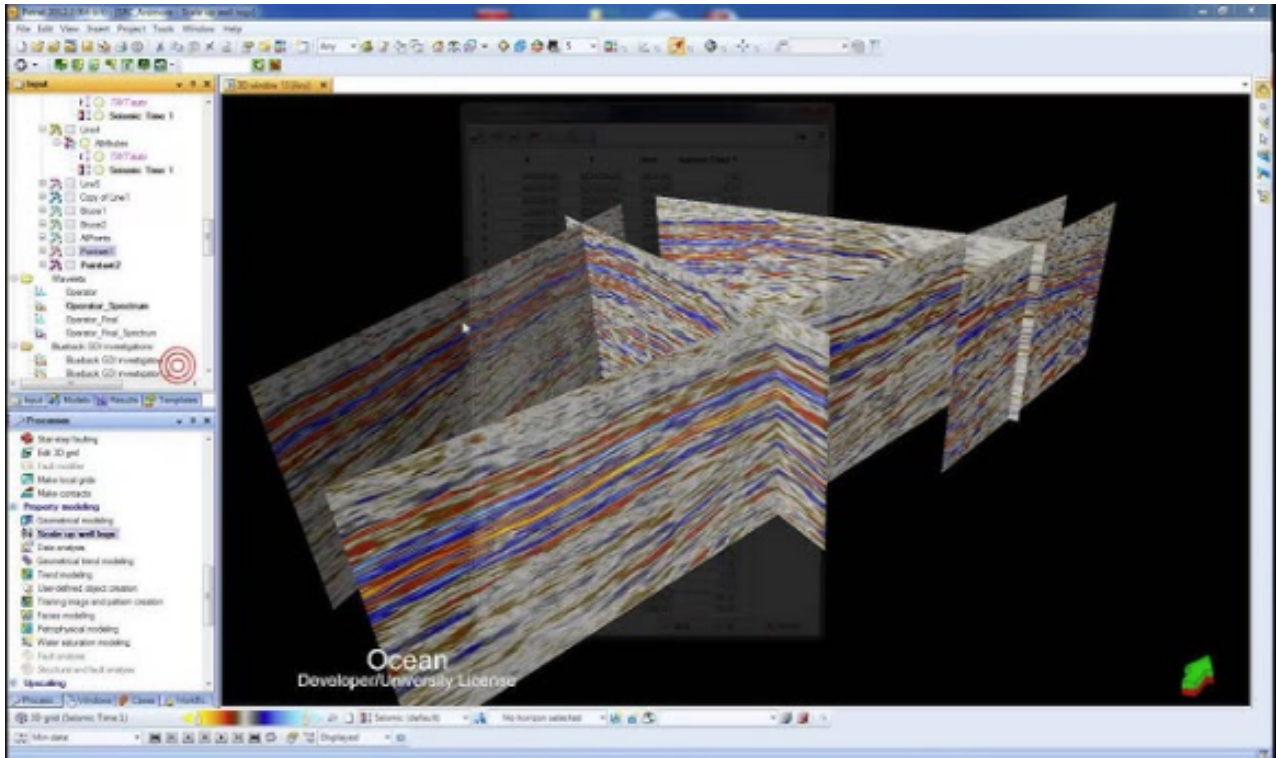


Figure 4. Several 2D seismic lines drawn together in 3D. The geologist sees how the subsurface looks along the different acquisition directions at once.

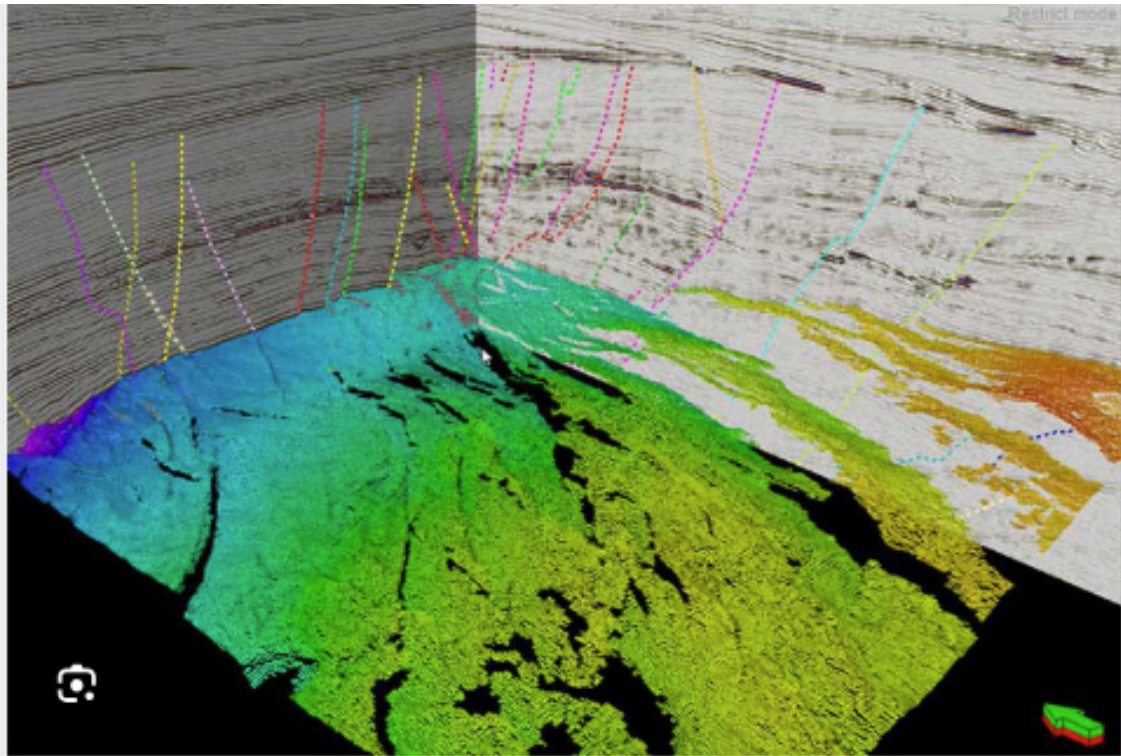


Figure 5. A reservoir model in 3D with a well trajectory through it. The well path against the body is visible at a glance.

2.4 Coordinate System

Petrel works in real-world coordinates: latitude/longitude or Easting/Northing. Reprojection is handled by the software, not by the user. Two datasets in different systems land in the right place without manual intervention. This is what makes the cross-view interactivity actually useful: the well in the map is the same well in the section and the same well in 3D, because all three views read from the same coordinate frame. Constra needs the same property for engineering data.

3. Reference Point: Surfer

Surfer (Golden Software) is a plotting and contouring package widely used in geoscience. It is included here as a different sort of reference. Petrel shows what a workspace can be. Surfer shows what plotting tools cannot do for someone who needs more than a finished figure. Surfer is good at what it does. It just is not a workspace, and a workspace is what engineering needs.

3.1 No dedicated views

Surfer does not have separate Map, Section, and 3D views that share state. Maps and sections both end up in the same generic plot window. There is no shared canvas, no automatic basemap, no cross-view synchronisation. Every figure is built by hand, and every figure is its own thing.

3.2 No project-level coordinate frame

Surfer does not put every dataset into a shared coordinate system. It plots data against whatever values are in the file, and it is up to the user to make sure those values agree across files. If two datasets are in different coordinate systems, the user notices that, not the software.

3.3 Manual workflow

Every layer, every object, every label has to be added by hand. That is fine for a one-off figure. It is not fine when many datasets have to stay consistent, when the underlying data changes regularly, or when the interpreter needs to switch between map, section, and 3D in the course of a working day.

Product page: <https://www.goldensoftware.com/products/surfer>

4. The Engineering-Geoscience Gap

The two products above are reference points, not rivals. The oil and gas industry has its workspace. Everyone, engineers included, has plotting tools. Engineering itself has no equivalent of either kind. The table below shows where each capability stands, in each industry, today.

Capability	Petrel (oil and gas)	Surfer (generic plotting)	Engineering geoscience today
Industry-specific integrated workspace	Yes	n/a	Nothing widely available
Single project, shared coordinate frame	Yes	No	Nothing widely available
Real-world coordinates, automatic reprojection	Yes	No	Handled per tool, per file
Linked Map, Section, and 3D views	Yes	No	Nothing widely available
Multiple simultaneous cross-sections	Yes	No	Manual in CAD
Native readers for the industry's primary data	Yes (oil and gas formats)	n/a	Fragmented across single-purpose tools
Cross-view selection	Yes	No	Nothing widely available

Constra is the rightmost column.

5. The Problem and Who Constrains Is For

5.1 The problem

An engineering project (a tunnel, a dam, a road cutting, a foundation, a slope) generates a lot of geological and geophysical data. Boreholes. In-situ tests like CPT and SPT. Surface geophysics: refraction, MASW, GPR, ERT. Topographic and bathymetric surveys. CAD plans. The engineer's own interpreted cross-sections.

Today that data is spread across separate tools. CAD for plans and sections. Surfer or similar for graphs. A GIS package for spatial layers. Excel for the tabular records that hold everything else. Nothing pulls these into one project the way Petrel does in oil and gas, and nothing keeps them in one coordinate system with linked map, section, and 3D views.

Constra exists to be that one thing.

5.2 Who Constrains is for

Constra is for the engineering geologist and the engineering geophysicist. The practitioner who interprets the near surface and then has to communicate that interpretation to civil designers, contractors, regulators, and clients. The fields of practice this covers:

- Geotechnical site investigation.
- Civil and construction engineering: foundations, slopes, dams, tunnels, roads, bridges, embankments.
- Mining engineering and exploration outside oil and gas.
- Environmental engineering and engineering hydrogeology.

What ties these together: the practitioner works in real-world coordinates, has several data types in every project, and has to produce interpreted outputs (cross-sections, maps, 3D models) that other people use to make engineering decisions.

5.3 What Constrains is not

- **Not a numerical solver.** Constra interprets and visualises. It does not run finite-element analyses, slope-stability calculations, groundwater models, or seismic processing pipelines. It produces interpreted outputs that those tools can read in.
- **Not a web app, at least not yet.** v0.1 is a Windows desktop application. The stack we picked could go elsewhere later, but the first release does not promise that.
- **Not a collaboration platform.** v0.1 is one practitioner working on one project at a time. Multi-user editing, cloud sync, and shared review come later.

6. Initial Release Scope (v0.1)

Constra is meant to be a real working tool, not a stripped-down proof of concept. The v0.1 scope below is what the first release has to do to be useful on an actual engineering project. Everything else can wait.

6.1 In scope for v0.1

- **A project.** A Constra project is a folder. Inside it: the project's data sources, its coordinate reference system, and a small metadata file.
- **Map View.** A 2D canvas drawing every dataset at its real location. Pan, zoom, layers on or off. The user draws cross-section corridors here (straight lines or polylines) for the Section View to follow.
- **Section View.** Vertical sections, several open at once, one per corridor on the map. Each section shows whatever data the corridor crosses (boreholes, in-situ test traces, surface geophysics) at the right depth and at the right distance along the line.
- **3D View.** The same data in 3D. Linked to the Map and Section views by a shared selection.
- **Data import.** Built-in readers for the formats engineering teams actually use. The final v0.1 list is still being decided. Current candidates: borehole records (AGS), in-situ tests (CPT, SPT), surface geophysics (refraction, MASW, GPR, ERT), DEM and DTM rasters (GeoTIFF, ASCII grid), GIS layers (shapefile, GeoJSON), CAD plans (DXF). SEG-Y is not on the list. Engineering geophysics is mostly not SEG-Y.
- **Coordinates.** Each project has one CRS, set when the project is created. Anything loaded into the project gets reprojected into that CRS by PROJ.
- **Cross-view selection.** Click a borehole, a station, or a feature in any view, and the same thing lights up in the other two.
- **Windows installer.** One MSI or exe for Windows 10/11, published on GitHub Releases.

6.2 Deferred past v0.1

The list below is what fits the engineering use case but is not in the first release. Things that belong to oil and gas (reservoir modelling, horizon picking on 3D seismic, SEG-Y attribute generation) are not deferred. They are simply not in scope.

- Editing or writing back to source files. v0.1 only reads.
- Multiple users on one project. Cloud sync. Shared review.
- Mac and Linux installers. The stack supports both, but v0.1 will not produce or support them.
- A Python scripting or plugin API.
- Web or SaaS deployment.
- Automatic report generation (PDF or DOCX export of interpreted sections and maps).

7. Technology Choice

7.1 Stack

- **Language:** Python 3.10 or newer.
- **GUI:** PySide6 (Qt 6).
- **3D and visualisation:** VTK, optionally through PyVista.
- **Coordinate systems:** pyproj, the Python binding for PROJ.
- **Raster I/O:** rasterio, for GeoTIFF and ASCII grid.
- **Vector I/O:** pyogrio or Fiona for shapefile and GeoJSON; geopandas for in-memory work.
- **CAD:** ezdxf for DXF.
- **Engineering parsers:** chosen during M1. Likely python-ags4 for AGS borehole data, plus dedicated or custom readers for CPT, refraction, MASW, GPR, ERT.
- **Windows packaging:** PyInstaller.
- **Tests:** pytest with pytest-qt.
- **Lint and format:** ruff, black.

7.2 Why this stack

Python with Qt and VTK is what ParaView and 3D Slicer are built on. Both are successful scientific visualisation applications, and both are open source. The hard parts of Constr's roadmap (large raster display, 3D scene management, coordinate reprojection, vector and CAD interop) all have established Python libraries.

Engineering-specific parsers, where they need to be written, are easier to write in Python than in anything else. Speed of iteration matters more right now than raw performance: getting the interaction model right is harder than making it fast. If a piece of the application later turns out to be a bottleneck, that piece can be rewritten in C++ or Rust without changing the rest.

7.3 Platform

v0.1 ships on Windows 10 and 11, because that is where engineering practitioners work. The stack itself is cross-platform, so Linux and macOS builds should be straightforward once the Windows build is solid. Neither is committed for v0.1.

8. Roadmap

Work is organised into milestones. Each milestone ends in a runnable application that does a little more than the one before it.

M0: Project bootstrap (complete)

- Public GitHub repo at github.com/atifali-pm/constra. MIT licence, README, and .gitignore in place.
- Python project skeleton: pyproject.toml, src layout, dev and docs extras.
- Ruff, black, pytest, and pytest-qt configured. GitHub Actions CI runs them on Python 3.10, 3.11, and 3.12.
- A PySide6 main window that launches, displays a placeholder, and closes cleanly.
- Docker development image with X11 forwarding so the app can be run visually on a Linux host.

M1: Project container, CRS, and engineering-format selection

- Define the on-disk project format: a folder plus a project.toml.
- Implement project create, open, save.
- Per-project CRS, set at project creation by EPSG code, stored in the project file.
- Wire pyproj so any loaded dataset is reprojected into the project CRS on the way in.
- Decide the v0.1 list of engineering data formats. Pick or write the parser for each.

M2: Map View

- Build the Map View widget (Qt Graphics View, or VTK in 2D mode).
- Load a DEM or DTM raster (GeoTIFF, ASCII grid) and draw it in the project CRS.
- Load a shapefile or GeoJSON vector layer.
- Load a CSV of point data, for example borehole heads, and label each point.
- Load a DXF plan via ezdxf and render it as a layer.
- Pan, zoom, layer visibility, and a side panel listing the layers.
- A tool that draws cross-section corridors (straight or polyline) on the map. The Section View will follow these corridors in M3.

M3: Section View (multiple sections)

- Build the Section View widget. Several sections can be open at once, one per corridor.
- For each section, project the data the corridor crosses (boreholes, in-situ test traces, surface geophysics) into (distance, depth) and render it.
- Configurable colour map, gain, vertical exaggeration, and depth scale.
- Link Section View to Map View. Selecting a corridor, a feature, or a borehole in one view highlights it in the other.

M4: 3D View

- Build the 3D View widget on VTK or PyVista.
- Render the basemap, boreholes, surface geophysics traces, and section corridors in 3D.
- Make the 3D View share the same selection model as the Map View and the Section View.

M5: Native engineering data and the first installer

- Implement the engineering-format parsers picked in M1: AGS borehole data, CPT/SPT, surface geophysics formats.
- Display borehole logs next to the section view.
- Package the application for Windows with PyInstaller.
- Publish a signed Windows installer to GitHub Releases.

When M5 ships, Construa v0.1 is a real workspace: one project, one coordinate system, linked Map, Section, and 3D views, and built-in readers for the data engineers actually use. It is the first usable release of the project, and what every later release is built on.